

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	---	---

Cloud and Aerosol Detection Software

Version 3

Product Specification

Version 1.3
07/07/2020

Reima Eresmaa
ECMWF

This documentation was developed within the context of the EUMETSAT Satellite Application Facility on Numerical Weather Prediction (NWP SAF), under the Cooperation Agreement dated 7 December 2016, between EUMETSAT and the Met Office, UK, by one or more partners within the NWP SAF. The partners in the NWP SAF are the Met Office, ECMWF, DWD and Météo France.

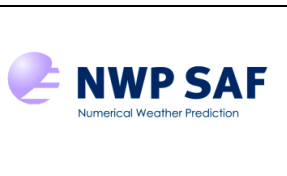
COPYRIGHT 2020, EUMETSAT, ALL RIGHTS RESERVED.

Change record			
Version	Date	Author / changed by	Remarks
1.0	10/03/2020	Reima Eresmaa	Initial version
1.1	07/04/2020	Reima Eresmaa	Include the required header information in the mandatory input. Clarify the aerosol type indexing.
1.2	06/07/2020	Reima Eresmaa	Include details on the simulated dataset used for MTG/IRS and IASI-NG data in CADS V2.3
1.3	07/07/2020	Reima Eresmaa	Include justification for using Fortran 90.

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
--	--	--

TABLE OF CONTENTS

1 INTRODUCTION	3
2 REQUIREMENT REVIEW	3
2.1 Development activity as planned in the NWP-SAF CDOP-3 proposal.....	3
2.2 User survey 2019	4
2.3 Summary	5
3 FUNCTIONALITIES.....	7
4 PERFORMANCE.....	8
5 LIMITATIONS	8
6 SYSTEM REQUIREMENTS	9
7 LIST OF PRODUCT REQUIREMENTS.....	9

 	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	--	---

1 INTRODUCTION

The Cloud and Aerosol Detection Software (CADS) is a flexible stand-alone cloud and aerosol detection package intended for radiance assimilation studies using high-spectral-resolution infrared (IR) sounders. The code was originally developed for the operational assimilation of AIRS (Atmospheric Infrared Sounder) radiances, and it has since been adapted to handle several other IR sounders. Regular updates are made to maintain the capability to reproduce performance characteristics as implemented in the ECMWF 4D-Var system. The software is applicable to both original and reconstructed radiance data.

Version 3.0 introduces the following new functionalities:

1. Distinguishing between Saharan dust, volcanic ash, and other aerosol
2. Channel-dependent flagging depending on the aerosol type
3. Situation-dependent identification of channels sensitive to underlying land surface
4. Detecting and flagging data affected by excessive Hydrogen Cyanide (HCN)
5. Processing capability for IKFS-2 (Infrared Fourier Spectrometer -2) sounder

2 REQUIREMENT REVIEW

The user requirements are reviewed against the NWP-SAF CDOP-3 proposal on the one hand, and as revealed by a recent user survey on the other.

2.1 Development activity as planned in the NWP-SAF CDOP-3 proposal

In addition to routine maintenance and responding to user feedback and help-desk requests, the NWP-SAF CDOP-3 project proposal outlines the following developments:

1. Upgrade software for prototype studies related to MTG/IRS & IASI-NG based upon simulated observations
2. Extend to high-spectral-resolution IR sounder data from Chinese satellites if available
3. Further exploitation of mapped imager products where available (e.g. CrIS/VIIRS)
4. Study algorithmic development to improve detection accuracy
5. Extend number of different aerosol species that can be discriminated
6. Develop ability to identify contaminated / uncontaminated channels depending on aerosol species and spectral signature

The planned activity item (1) was included in CADS V2.3, that was released in January 2017. The simulated MTG/IRS and IASI-NG observation dataset, that was used in preparation of V2.3, was produced locally at ECMWF by applying RTTOV 11.3 on a 137-level diverse profile dataset. The planned activity item (2) was implemented in version V2.4, that was released in March 2019. V2.4 also includes the ability to identify aerosol-contaminated channels (in relation to the activity item (6)); however, this implementation

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	---	--

does not take the aerosol species into account. Requirement remains for completing the activity items (3), (5), and (6). As for item (4), the algorithmic development is continued, and this work may result in a software change in a future version of CADS.

2.2 User survey 2019

To obtain an up-to-date view on user requests, a survey was carried out in 2019 Q4. The survey consisted of a questionnaire gauging respondents' familiarity with existing software versions, their current use of satellite-based infrared sounder data, and their wishes regarding new functional elements to be implemented in the short term (i.e., by the end of NWP-SAF CDOP-3, February 2022) and in a longer time perspective. The survey was distributed to potential users in the International TOVS Study Conference 22 (ITSC-22; 31 October – 6 November 2019, Saint-Sauveur, Quebec, Canada) and it was specifically advertised during the NWP Working Group meeting at the conference. The survey was also distributed to registered CADS users by e-mail in November, calling for responses by the end of year. Finally, the survey was put accessible through the NWP-SAF web site and it was advertised on the NWP-SAF Twitter account.

At the time of closing, the survey at the web site had produced eight responses, while four additional responses were received by either e-mail or in hard copies as returned during the ITSC-22 conference. The total number of responses was 12, and they are all considered anonymous here. The results are collected in Table I and they are discussed below.

Responses on Q1 indicate that four out of 12 respondents are occasional users, while the other eight respondents have not yet tried using CADS.

Regarding respondents' current use of different IR sounders (Q2), it is summarized that majority of the respondents are working either on well-established hyper-spectral sounders (IASI, CrIS and AIRS) or on the future European sounders (IASI-NG and MTG-IRS). Users of Chinese and Russian sounders, and those of HIRS, are in minority in this sample. It is worth noting that two respondents indicated use of IRIS (flown in 1970's on NIMBUS satellites).

From replies on Q3, it is concluded that the respondents consider prompt implementation of aerosol type recognition as a priority. Similarly, channel-specific flagging for land-sensitivity is generally considered desirable by users. There is less imminent interest towards the HCN detection or processing capability for IKFS-2. Of the two responses on "Other", one requested clearer guidance on how to include IR sounders and/or collocated imagers that are not supported by default. The other expressed interest on including the processing capability for IRIS.

Responses to Q4 are considered as a strong indication of users being interested on accounting for different cloud types in future versions of CADS. Similarly, the users are very keen to make improved use of auxiliary information and to being provided with advanced tools for studying output diagnostics. Consequently, these three lines of development will be given high priority when planning future lines of development. The

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	---	---

other provided options have each gathered some interest, and they will remain candidates for further development, but with a somewhat lower priority. Again, it is worth noting the one vote on “Other” specifying a desire for better integration with other NWP-SAF tools and their user interfaces.

2.3 Summary

In the short term, users have indicated a considerable interest towards the implementation of aerosol type recognition. As this item was also requested as part of the NWP-SAF CDOP-3 proposal, this will make a natural headline item for CADS V3.0. To comply with the CDOP-3 proposal, the recognition of aerosol type will also have implications on the way contaminated and uncontaminated channels are separated from each other. As the user survey demonstrates considerable interest on the flagging for land-sensitivity, as well as some interest on the excessive HCN detection and processing capability for IKFS-2, also these functionalities will be included in V3.0. Additionally, emphasis will be put on clarifying the guidance on how to include new observation types that are not supported by default.

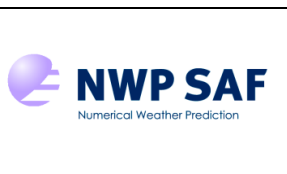
 	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	--	---

Table I: User survey 2019.

Q1: Are you familiar with CADS?					
I am a frequent user of CADS				No responses	
I am an occasional user of CADS				4 responses	
I have heard of CADS but never used it				7 responses	
I have never heard of CADS before				1 response	
Q2: Which IR sounder data do you use at your work (please select all that apply)?					
HIRS	3 responses	IKFS-2	1 response	IASI-NG	7 responses
AIRS	5 responses	HIRAS	2 responses	MTG-IRS	4 responses
IASI	10 responses	GIIRS	2 responses	Other	2 responses
CrIS	6 responses				
Q3: Which software improvements would you like to see included in the next release of CADS?					
Processing capability for IKFS-2				2 responses	
Aerosol type recognition				9 responses	
Detection of excessive Hydrogen Cyanide (HCN)				3 responses	
Flagging for land-sensitivity				6 responses	
Other				2 responses	
Q4: Which aspects would you prefer seeing developed further in the context of CADS?					
Accounting for different cloud types (e.g. ice / liquid / mixed phase)				8 responses	
Accounting for more aerosol species (other than volcanic ash and Saharan dust)				4 responses	
Accounting for more trace gases (other than HCN)				4 responses	
Accounting for more surface types (other than just land/sea)				3 responses	
Including new sounders				4 responses	
Making better use of auxiliary information (such as collocated imagers)				6 responses	
Improving the consistency with observation and background error statistics				3 responses	
Including physically-based constraints				2 responses	
Using a priori cloud information				1 response	
Developing tools for output diagnostics				6 responses	
Other				1 response	

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	---	---

3 FUNCTIONALITIES

CADS produces situation- and channel-specific flagging for infrared radiance data. Flags are raised for the following data events:

- Cloud contamination
- Aerosol contamination
- Contamination by excessive trace gas concentration
- Sensitivity to underlying land surface

Running the software requires the following mandatory input information:

- List of sounder channel indices
- Observed and calculated (from a short-range forecast) brightness temperatures for the required instrument channels
- Relative height assignments for the required instrument channels
- Height assignments corresponding to the tropopause and top of planetary boundary layer at each location
- Fraction of land in the sounder's field-of-view

The following input information is optional and made use of whenever they are given:

- Namelist files specifying the desired configuration for the cloud, aerosol, trace gas, and land sensitivity detection
- List of used channels for collocated imager statistics
- Fractional coverage of each cluster of imager pixels within sounder's field-of-view (FOV)
- Mean observed brightness temperature in each cluster for each imager channel
- Standard deviation of observed brightness temperature in the FOV for each imager channel
- Calculated (from a short-range forecast) brightness temperature for each imager channel

The output consists of flags indicating cloud contamination, aerosol contamination, trace-gas contamination, and land sensitivity on each input channel. The flags are binary such that unaffected channels are set to flag=0 while those affected are set to flag=1. Additionally, index of aerosol species is output (0=No aerosol, 1=Saharan dust, 2=Volcanic ash, 3=Other aerosol, 4=Any aerosol over land).

Table II: Provision of meaningful default settings for different IR sounders.

	AIRS	IASI	CrIS	MTG-IRS	IASI-NG	HIRAS	GIIRS	IKFS-2
Cloud contamination								
Aerosol contamination								
Trace-gas contamination								
Land sensitivity								

Default settings for various sounder/flag combinations are provided as shown in Table II. Green-shaded cells indicate combinations supported by including meaningful defaults. On MTG-IRS, IASI-NG, HIRAS, GIIRS, and IKFS-2, no meaningful defaults are given for the aerosol- and trace-gas contamination parameters: in these cases, default settings will make the detection inactive. This is due to the prevailing lack of access to extensive test datasets that are required for finding sufficiently robust detection parameters.

4 PERFORMANCE

The calculation of cloud, aerosol, trace-gas, and land-sensitivity flags is performed efficiently, such that the required computing resources are only a small fraction of the total used in the radiance assimilation. Specifically, this requires that the CPU time required for production of the flags is small in comparison with the forward model calculation. The efficient performance is facilitated by making use of quantities that are readily available in a data assimilation system.

5 LIMITATIONS

The ability of the software to detect cloud, aerosol, and trace gases is limited to the information available either in the observed radiances, in the departures of observed radiances from their calculated counterparts, or in collocated imager radiance statistics. As with any detection scheme, there will inevitably be cases of missed detections and false positives. However, given that the software is by design made consistent with the infrared sounder assimilation at ECMWF, the operational use of these methods suggests that the radiative effect associated with such incorrect flagging is likely to be small. The infrared radiance assimilation is repeatedly demonstrated to produce a beneficial short- and medium-range forecast impact.

Default settings are provided for application to AIRS, IASI, CrIS, MTG-IRS, IASI-NG, HIRAS, GIIRS, and IKFS-2 radiances. Extending the processing capability to other

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-014 Version : v1.3 Date : 07/07/2020
---	---	--

sounders is technically possible, but it will require substantial modifications to source code. The user guide contains detailed instructions on how to do this.

6 SYSTEM REQUIREMENTS

The software operates on UNIX-based computing platforms and it is provided as a set of shell scripts and source files written in FORTRAN 90 programming language. A pre-installed FORTRAN 90 compiler is a mandatory requirement. The user may need to modify the provided makefile to correspond the desired compiler and/or their local computing platform.

The software is written in FORTRAN 90 programming language to facilitate easy implementation in most NWP system architectures.

Installing and running the provided test scripts requires 100MB of free memory.

Running the software requires no access to external interfaces. However, the user needs to be able to retrieve observed brightness temperature data (e.g. from BUFR or ODB files) and to calculate brightness temperature and channel-specific height assignments (e.g. by applying the RTTOV radiative transfer software on NWP model output) to provide input for the software.

7 LIST OF PRODUCT REQUIREMENTS

1. The software compiles, runs and produces similar output on a variety of Unix platforms.
2. The software includes test scripts and sample data representative of AIRS, IASI, CrIS, HIRAS, GIIRS, IKFS-2, MTG-IRS, and IASI-NG sounders.
3. The software identifies cloud-contaminated IR channels at each location.
4. The software identifies locations where aerosol contamination occurs and distinguishes between Saharan dust, volcanic ash, and other aerosol.
5. Depending on aerosol species, the software identifies contaminated IR channels at each location.
6. The software identifies locations and IR channels affected by anomalously high concentration of atmospheric HCN.
7. The software identifies IR channels sensitive to underlying surface at each location.
8. Production of the cloud, aerosol, trace-gas, and land sensitivity flags is efficient and consumes only a small fraction of computing time as compared with forward modelling of brightness temperatures.
9. The output flagging is consistent with performance of the ECMWF 4D-Var data assimilation system.